RetailBot Pro: A Framework for Resource-Efficient, Domain-Adapted Conversational AI in Retail via QLoRA Fine-Tuning

Author: Ruben Stanley John Benedict Lobo

GitHub:

https://github.com/rubenslobo/Applied-Thesis-Project---Conversational-AI---Research-Paper

I. ABSTRACT

- The rise of Large Language Models (LLMs) has opened new avenues for building domain-specific conversational systems that offer precise, context-aware insights within specialized sectors. This paper introduces RetailBot Pro, a comprehensive conversational AI tailored specifically for the retail and e-commerce industry. Our approach leverages Quantized Low-Rank Adaptation (QLoRA) — used to fine-tune the TinyLlama-1.1B-Chat model — on a structured retail customer support dataset sourced from HuggingFace.
- To fine-tune the model, we utilized the Bitext customer sup-port LLM chatbot training dataset and applied 4-bit NF4 quantization, successfully conducting training on consumer-grade hardware (Google Colab T4 GPU). The fine-tuned model is deployed as an interactive Gradio chatbot capable of handling retail queries including order tracking, return procedures, refund policies, and shipping enquiries. Exper-imental results indicate that this domain-adapted approach results in high response accuracy, minimal hallucination, and sub-3-second latency.

II. INTRODUCTION

2.1 Background and Motivation

- The evolution of transformer-based architectures has re-defined the boundaries of human-computer interaction in commercial settings. Modern conversational agents have moved beyond the limitations of rigid, rule-based logic to leverage Large Language Models capable of maintaining fluent dialogue across diverse customer scenarios.
- However, in the field of retail and e-commerce, general-purpose models often struggle with “hallucinations” — gen-erating confident but factually incorrect responses when faced with specialized retail queries such as order status, return eligibility, or refund policies (Ji et al., 2023).
- The retail industry requires a higher standard of precision. Customers require immediate, accurate information regard-ing their purchases, shipping, and service entitlements. The mismatch between the generalized capabilities of off-the-shelf LLMs and the rigorous demands of retail stakeholders necessitates the creation of specialized systems that anchor generative fluency to verified, domain-specific knowledge.

2.2 Research Problem

- Current retail AI tools generally fall into two categories: generic LLM interfaces that lack deep domain expertise, or static FAQ systems that lack conversational depth. Nei-ther fully meets the needs of retail customers who require nuanced, contextually grounded explanations. Addition-ally, the massive computational overhead required to fully fine-tune state-of-the-art models often makes specialized development inaccessible for small-scale businesses.
- This study seeks to bridge this gap by addressing a core technical question: Is it possible to build a high-performing, retail-relevant AI system on commodity hardware by lever-aging QLoRA-based parameter-efficient fine-tuning on a structured customer support dataset?

2.3 Research Objectives

- To address this problem, the study focuses on the following three objectives:
  Pipeline Development & Evaluation: To build and assess a QLoRA-based fine-tuning workflow for the TinyLlama-1.1B-Chat model using a specialized retail customer support dataset, demonstrating that effective domain adaptation is possible on limited hardware.
  - **Hallucination Reduction:** To implement structured prompt engineering and low-temperature generation strategies that minimize hallucination and improve fac-tual accuracy in retail-specific responses.
  - **System Deployment:** To engineer an interactive conver-sational interface via Gradio capable of handling real-world retail queries in a production-like environment.

2.4 Scope and Significance

The scope of this research covers the entire development cycle of RetailBot Pro, ranging from dataset curation and model fine-tuning to the final deployment of the chatbot. The system addresses a wide spectrum of retail topics, including order management, return and refund procedures, shipping enquiries, and product support.

Beyond its immediate application, this work is significant because it provides a reproducible, cost-effective blueprint for building domain-specific AI. This framework can be adapted by researchers in other customer-facing industries such as logistics, hospitality, or banking, where precision is critical but computing budgets are limited.

III. INDUSTRY ANALYSIS

3.1 Overview of the Retail and E-Commerce Landscape

The global e-commerce market is experiencing unprecedented growth, with its valuation expected to surpass $7.9 trillion by 2027. AI-driven customer support has emerged as a primary driver for operational efficiency and customer satisfaction. As retail delivery shifts toward digital-first environments, there is a growing demand for tools that go beyond static FAQs or scripted chatbots. Traditional support structures are no longer sufficient for modern online shoppers who expect immediate, accurate, and personalized responses at any hour of the day.

3.2 AI Adoption Challenges in Retail

While the potential for AI in retail customer support is high, several hurdles prevent seamless adoption:

- **The Risk of Hallucination:** General-purpose models often generate incorrect product or policy claims, which can mislead customers and erode trust in the brand's digital presence.
- **Specialized Vocabulary:** Retail discourse is rich with domain-specific terms — such as SKU identifiers, SLA windows, RMA procedures, and chargeback policies — that models must be specifically tuned to understand and apply accurately.
- **Hardware Barriers:** The high cost of training large-scale models often excludes smaller retailers. Without access to massive cloud budgets, many organizations are locked out of custom AI development.
- **Lack of Continuity:** Retail support often requires multi-turn dialogue. Many basic systems lack session coherence, making them unable to track the context of an evolving customer complaint.

3.3 The Rise of AI-Augmented Retail Tools

Recent breakthroughs in Parameter-Efficient Fine-Tuning (PEFT) — specifically LoRA and QLoRA — have democra-tized AI development for specialized domains. These tech-niques allow developers to customize billion-parameter mod-els using modest hardware without sacrificing conversational quality. For the retail sector, this means the vision of an always-on, expert-level digital support agent is finally becom-ing a practical, cost-efficient reality.

3.4 Business Value and Industry Requirements

The implementation of domain-specific retail AI provides measurable advantages across several key areas:

- **24/7 Customer Support:** Continuous access to high-quality retail assistance without human intervention, re-ducing average handle time and support costs.
- **Operational Efficiency:** A significant reduction in the administrative burden on human support agents, who are

often overwhelmed by repetitive policy or order queries.

- **Global Scalability:** The ability to provide high-quality, standardized support to customers across diverse geogra-phies and time zones.
- **Brand Consistency:** Delivery of consistent, policy-aligned responses that reduce the risk of misinformation and protect brand reputation.

For retail organizations and e-commerce platforms, these sys-tems are more than just “chatbots”; they represent a fundamen-tal upgrade to the infrastructure of customer service delivery.

IV. LITERATURE REVIEW

4.1 Overview

Since the landmark introduction of transformer architectures by Vaswani et al. (2017), the body of research regarding specialized conversational AI has grown exponentially. This review synthesizes current scholarship across three critical areas: Parameter-Efficient Fine-Tuning (PEFT) techniques, hallucination mitigation strategies, and the practical applica-tion of AI within customer support environments.

4.2 Parameter-Efficient Fine-Tuning (PEFT)

To address the challenges of model adaptation, Hu et al. (2022) introduced Low-Rank Adaptation (LoRA). Rather than retrain-ing all model parameters, LoRA freezes the original weights and inserts trainable rank-decomposition matrices into the transformer layers. The authors proved that this method could match the performance of full fine-tuning while requiring only a fraction of the trainable parameters. This is a corner-stone of the RetailBot Pro architecture; we employ LoRA adapters (rank 16) across the query and value projection layers (q proj, v proj) to achieve deep domain adaptation within the TinyLlama-1.1B model.

4.3 Quantized Fine-Tuning

Building upon the LoRA framework, Dettmers et al. (2023) developed QLoRA, which allows for the fine-tuning of mod-els using 4-bit NormalFloat (NF4) quantization. This break-through demonstrated that even massive 65B-parameter mod-els could be tuned on a single 48GB GPU without losing performance. For this research, QLoRA is the primary tech-nical enabler. By compressing the TinyLlama-1.1B memory footprint to approximately 1.1 GB, we successfully conducted fine-tuning within the hardware constraints of a Google Colab T4 GPU, proving that sophisticated retail AI development is possible on consumer-level resources.

4.4 Hallucination in Generative Models

Ji et al. (2023) conducted a comprehensive survey of hallucina-tion in natural language generation, categorizing it into intrin-sic and extrinsic types. Their findings underscore the critical importance of domain grounding and controlled generation pa-rameters in high-stakes applications. RetailBot Pro addresses this through structured prompt engineering, low-temperature sampling (0.3), repetition penalties, and fine-tuning on cu-rated instruction-response pairs reflecting real retail support interactions.

4.5 Existing Frameworks and Tools

Tool / Frame-work	Role in RetailBot Pro
HuggingFace Transformers	Model loading, tokenizer, BitsAndBytesConfig
PEFT (Hu et al., 2022)	LoRA adapter configuration and model wrapping
TRL / Train-ingArguments	Supervisedfine-tuning pipeline management
Bitext CS Dataset	Primary domain-specific training corpus
Gradio	Interactive chatbot frontend with public share
Google Colab T4 GPU	Hardware platform for model training

Table 1: Frameworks and Tools in RetailBot Pro

4.6 Gaps in Existing Literature

Despite extensive research on general-purpose LLMs and parameter-efficient tuning, several significant gaps remain in the retail AI space. First, most existing chatbot research focuses on general dialogue systems rather than the specific nuances of retail customer support terminology and policy structures. Second, few studies have demonstrated a complete QLoRA-based fine-tuning pipeline on a structured customer support dataset deployable on commodity hardware. RetailBot Pro aims to bridge these gaps through a full-stack implemen-tation and subsequent empirical analysis.

4.7 Contribution of This Study

This research provides a complete, reproducible, and hardware-accessible framework for domain-specific retail AI. Our pri-mary contributions include:

- **A Domain Dataset:** A structured instruction-following corpus derived from the Bitext customer support dataset, formatted for retail dialogue fine-tuning.
- **Validated Pipeline:** A proven QLoRA fine-tuning work-flow for TinyLlama-1.1B optimized for T4 GPU hardware.
- **Deployment Interface:** A production-ready Gradio chat-bot supporting real-time retail query resolution with inte-grated hallucination mitigation.
- **Empirical Data:** A comprehensive evaluation of the sys-tem's factual accuracy, response speed, and dialogue coher-ence on retail test scenarios.

V. METHODOLOGY

The development of RetailBot Pro follows a structured four-stage pipeline: dataset construction, model fine-tuning, hallu-cination mitigation, and Gradio deployment.

5.1 Dataset Construction

The training corpus was sourced from the publicly avail-able **bitext/Bitext-customer-support-llm-chatbot-training-dataset** on HuggingFace. This dataset contains structured instruction-response pairs covering a comprehensive range of retail customer support scenarios, including order manage-ment, return policies, shipping enquiries, refund procedures, and account management.

All training data is formatted using a consistent instruction-response prompt template:

```
### Instruction:
{instruction}

### Response:
{response}
```

To ensure training efficiency and prevent memory overrun on the T4 GPU, the dataset was sampled to **2,000 high-quality instruction-response pairs**, sufficient to achieve meaningful domain adaptation for the core retail support use cases.

5.2 QLoRA Fine-Tuning Configuration

We utilized **TinyLlama/TinyLlama-1.1B-Chat-v1.0** as the base model, loaded via BitsAndBytesConfig using 4-bit NF4 quantization with fp16 compute data type and double quan-tization enabled. This successfully compressed the model's VRAM requirements, making training feasible on the Google Colab T4 GPU (15 GB VRAM).

LoRA adapters are configured with rank r=16, scaling factor alpha=32 (alpha/rank = 2.0), and dropout rate 0.1, targeting the query and value projection matrices (q proj, v proj). This targeted adapter approach minimizes the number of trainable parameters while focusing adaptation on the attention layers most critical for dialogue quality.

5.2.1 Training Execution

Training was performed using the `TrainingArguments` and `Trainer` API from the HuggingFace Transformers li-brary. Key hyperparameters were configured as follows: 2 training epochs, per-device batch size of 4, gradient accumu-lation steps of 4 (effective batch size of 16), learning rate of 2e-4, fp16 mixed-precision training, and paged AdamW 8-bit optimizer. The model was saved at the end of each epoch us-ing `save strategy='epoch'`, and the complete training process on 2,000 samples was completed in approximately 20 minutes on the T4 GPU.

5.3 Hallucination Mitigation Strategy

RetailBot Pro implements a multi-layered hallucination mit-igation strategy during inference. First, the system prompt explicitly constrains the model to its retail support role: *"You are a professional retail customer support assistant. Provide clear, polite and factual answers only."* Second, generation is controlled with low temperature (0.3), top-p sampling (0.9), repetition penalty (1.2), and a maximum new token limit of 150 tokens. These constraints collectively steer the model toward factual, concise, policy-aligned responses.

5.4 Deployment Interface

For the user-facing component, we developed a Gradio `ChatInterface` with a soft theme, a descriptive title, and four pre-configured retail example queries covering the most common support scenarios: order tracking, returns, shipping address changes, and refund policy enquiries. The interface is deployed via a publicly accessible Gradio share link, facilitat-ing real-time testing and stakeholder demonstration.

VI. EXPERIMENTS AND RESULTS

6.1 Evaluation Objectives

The assessment of RetailBot Pro focused on five key perfor-mance indicators:

1. **Factual Accuracy:** Alignment with established retail policies and order management standards.
2. **Response Relevance:** The appropriateness and speci-ficity of generated responses to retail queries.
3. **Response Latency:** The end-to-end time from query submission to generation.
4. **Dialogue Coherence:** The system's ability to maintain context across multi-turn exchanges.
5. **Hallucination Mitigation:** The frequency of factually unsupported claims within the retail domain.

6.2 Experimental Setup

Testing was conducted using the Gradio interface deployed on Google Colab. We reloaded the fine-tuned TinyLlama-1.1B adapter and evaluated the system on a curated test set of 50 queries categorized into:

- Order Management (e.g., order status, cancellation).
- Returns and Exchanges (e.g., return eligibility, exchange process).
- Refund Policies (e.g., refund timelines, partial refunds).
- Shipping and Delivery (e.g., address changes, delivery de-lays).
- Account and Payment (e.g., payment methods, billing queries).

6.3 Performance Metrics

6.4 Sample Query-Response Evaluations

6.5 Training Convergence

Training loss decreased consistently across the 2-epoch train-ing window, from approximately 2.61 at the initial logging step to 1.43 at the final training step, representing a 45.2% reduction. No significant overfitting was observed, consis-tent with the regularization provided by LoRA dropout (0.1) and the paged AdamW optimizer. Model checkpoints were

Metric	Description	Result
Response Accuracy	Factual correctness rated against retail policies	88% (44/50 correct)
Response Relevance	Proportion of on-topic responses by evaluators	90%
Response Latency	Avg wall-clock time from query to response	1.5–2.8 seconds
Hallucination Rate	Proportion of responses with fabricated claims	8% (4/50)
Multi-turn Coherence	Follow-up queries resolved correctly	84% (21/25)
User Satisfaction	Subjective rating 1–5 by 8 evaluators	4.3 / 5.0 avg

Table 2: RetailBot Pro Evaluation Results

saved at each epoch, and the best-performing checkpoint was retained for evaluation and deployment.

6.6 Limitations Observed

Several system limitations were identified during evaluation. The current implementation does not incorporate a persis-tent knowledge retrieval layer (such as RAG), which means the model relies entirely on its fine-tuned parametric knowl-edge for responses. This makes it susceptible to hallucination for highly specific queries — such as exact return window durations for specific product categories — not adequately represented in the training corpus. Additionally, queries using highly informal phrasing occasionally produced suboptimal responses when colloquial terms did not align with the formal vocabulary of the training data. Future iterations should in-tegrate a FAISS-based RAG pipeline to ground responses in verified policy documents.

6.7 Summary of Results

The experimental results collectively affirm that RetailBot Pro delivers reliable, retail-grounded conversational responses at production-grade latency. The combination of QLoRA fine-tuning for domain vocabulary adaptation and structured prompt engineering for hallucination control produces a sys-tem that substantially outperforms generic LLM responses on retail support content, achieves sub-3-second response latency suitable for real-time customer support, and maintains conver-sational coherence across multi-turn retail dialogue sequences.

VII. DISCUSSION

The findings of this study yield several insights relevant to the design of domain-specific conversational AI systems for retail. First, QLoRA fine-tuning on a structured, domain-specific instruction dataset produces qualitatively superior re-tail support responses compared to prompting a generic base

Query	Outcome
Where is my order?	PASS – Policy-aligned response with tracking steps
How do I return a product?	PASS – Complete return procedure with label instructions
Can I change my shipping address?	PASS – Accurate response on change window and escalation
What is your refund policy?	PASS – Refund timelines and eligibility correctly described
How long does shipping take?	PASS – Standard and express timeframes correctly stated
Can you process my refund now?	PASS – Correctly deferred to escalation without fabricating confirmation

Table 3: Sample Query Evaluation Results

model. The adaptation process enables the model to internal-ize retail-specific vocabulary, policy structures, and response conventions that are not adequately represented in general pre-training corpora.

Second, the choice of LoRA target modules has a measurable impact on domain adaptation quality. Targeting q proj and v proj — the query and value projection matrices — focuses adapter learning on the attention mechanisms most responsible for contextual response generation. Preliminary experiments with only q proj adapters produced noticeably less coherent policy-related responses, confirming the value of including v proj in the adapter configuration.

Third, the hallucination mitigation strategy — combining structured system prompts with low-temperature sampling and repetition penalties — demonstrably reduces the frequency of fabricated claims in retail support scenarios. However, the residual 8% hallucination rate underscores the continued importance of knowledge grounding. Future work should inte-grate a Retrieval-Augmented Generation (RAG) pipeline using FAISS vector search over retailer-specific policy documents to further reduce this rate.

From an industry perspective, RetailBot Pro demonstrates that high-quality domain-specific conversational AI is achievable with accessible hardware and open-source tooling. The com-plete training and deployment pipeline executes on Google Colab's free T4 GPU tier in under 30 minutes, establishing a reproducible template deployable by retailers, e-commerce platforms, and customer support teams without specialized ML infrastructure.

Ethical considerations for production deployment include the risk of embedding policy inaccuracies present in the training corpus, the need for regular model updates to reflect evolving return and refund policies, and the importance of transparent escalation pathways to human agents for complex or sensi-tive customer situations. Future work should also investigate persistent session memory architectures, multilingual support for global customer bases, and integration with e-commerce platforms via API connectors for real-time order data access.

VIII. CONCLUSION

This research presented RetailBot Pro, a domain-specific con-versational AI system for the retail and e-commerce industry, developed through QLoRA fine-tuning of TinyLlama-1.1B-Chat on the Bitext customer support dataset. The system was trained using 4-bit NF4 quantization on a commodity T4 GPU and deployed as an interactive Gradio chatbot with integrated hallucination mitigation strategies.

Experimental evaluation across 50 structured test queries demonstrated 88% response accuracy, 90% response relevance, sub-3-second latency, 8% hallucination rate, and 84% multi-turn coherence, collectively establishing RetailBot Pro as a performant and practically deployable retail conversational AI. The results confirm that parameter-efficient fine-tuning on structured domain data constitutes an effective and accessible paradigm for domain-specific AI development, with direct ap-plicability to retail customer support, e-commerce operations, and post-purchase service management.

Future research directions include integration of RAG for pol-icy document grounding, persistent cross-session memory, multilingual customer support, voice interface integration, and fine-tuning on live support log data for further domain spe-cialization. The blueprint established in this work is directly transferable to analogous customer-facing domains including logistics, hospitality, banking, and telecommunications.

APPENDIX

A. System Architecture Summary

RetailBot Pro integrates the following open-source components: fine-tuning via HuggingFace Transformers, PEFT, and TRL; generative inference via the fine-tuned TinyLlama adapter; and conversational UI via Gradio ChatInterface.

Component	Specification
Base Model	TinyLlama/TinyLlama-1.1B-Chat-v1.0
Quantization	4-bit NF4, fp16 compute, double quantization
LoRA Configuration	r=16, alpha=32, dropout=0.1, q proj + v proj
Training Epochs	2 (effective batch 16, LR 2e-4, paged AdamW 8-bit)
Training Dataset	Bitext customer-support-llm-chatbot (2,000 samples)
Generative LLM	Fine-tuned TinyLlama-1.1B adapter
Generation Config	temp=0.3, top p=0.9, rep penalty=1.2, max new tokens=150
Chatbot Interface	Gradio ChatInterface with retail example queries
Training Platform	Google Colab T4 GPU (15 GB VRAM)
GitHub Repository	`https://github.com/rubenslobo/Applied-Thesis-Project---Conversational-AI---Research-Paper`

TABLE IV: RetailBot Pro System Specification

B. Training Configuration Code

```
training_args =
   TrainingArguments(
   output_dir="./retail-
   llm",
   per_device_train_batc
   h_size=4,
   gradient_accumulation
   _steps=4,
```

```
    num_train_epochs=2,
    learning_rate=2e-4,
    fp16=True,
    logging_steps=50,
    save_strategy="epoc
    h",
    optim="paged_adamw_
    8bit",
    dataloader_pin_memo
    ry=False
)
```

REFERENCES

[1] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008. `https://arxiv.org/abs/1706.03762`

[2] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2022). LoRA: Low-rank adaptation of large language models. *International Conference on Learning Representations*. `https://arxiv.org/abs/2106.09685`

[3] Dettmers, T., Pagnoni, A., Holtzman, A., & Zettlemoyer, L. (2023). QLoRA: Efficient finetuning of quantized LLMs. *Advances in Neural Information Processing Systems*, 36. `https://arxiv.org/abs/2305.14314`

[4] Zhang, P., Zeng, G., Wang, T., & Lu, W. (2024). TinyLlama: An open-source small language model. *arXiv preprint arXiv:2401.02385*. `https://arxiv.org/abs/2401.02385`

[5] Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Zeng, E., Yeh, I. W., Joty, S., & Faltings, B. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 1–38. `https://arxiv.org/abs/2202.03629`

[6] Bitext. (2023). Bitext customer support LLM chatbot training dataset. HuggingFace Hub. `https://huggingface.co/datasets/bitext/Bitext-customer-support-llm-chatbot-training-dataset`

[7] Hugging Face. (2024). PEFT: State-of-the-art parameter-efficient fine-tuning library. `https://huggingface.co/docs/peft/`

[8] Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., & Rush, A. M. (2020). HuggingFace's transformers: State-of-the-art natural language processing. *EMNLP 2020*. `https://arxiv.org/abs/1910.03771`

Authors: Ruben Stanley John Benedict Lobo

Acknowledgement

The authors acknowledge the open-source contributions of the HuggingFace, PEFT, TRL, and Gradio communities whose frameworks and tools form the technical foundation of RetailBot Pro. The Bitext customer-support-llm-chatbot-training-dataset, sourced from the HuggingFace Hub, provided the primary training corpus for this research.